StockSense AI

Retail demand forecasting → practical stock decisions

VUT AI SOLUTIONS



The retailer’s stock problem



TOO LITTLE

Empty shelves and missed demand

TOO MUCH

Cash tied up in slow-moving stock

Objectives and success criteria

01

7 dated forecasts

for every selected product

02

Lower daily MAE

than the weekly benchmark

03

Clear stock logic

with invalid-input handling

04

24 / 25

supported requests recognised
Recorded curated test

The dataset and its four fields



913,000

rows · 2013–2017 · 10 stores · 50 items

Selected subset

9,130 rows

DATE · STORE · ITEM · SALES

One store, five items, seven days.
No inventory or cost records.

Training, validation and final testing

2013–2015	2016	2017
TRAIN Fit candidate models	VALIDATE Choose by daily MAE	TEST Frozen final evaluation

The three forecasting methods

WEEKLY NAIVE

Repeat last week

Benchmark

RANDOM FOREST

Many decision trees

Learns recent + calendar patterns

LSTM

Sequence neural network

Learns from 28-day windows

The LSTM deep-learning comparison

28 daily sales

Scale from past only

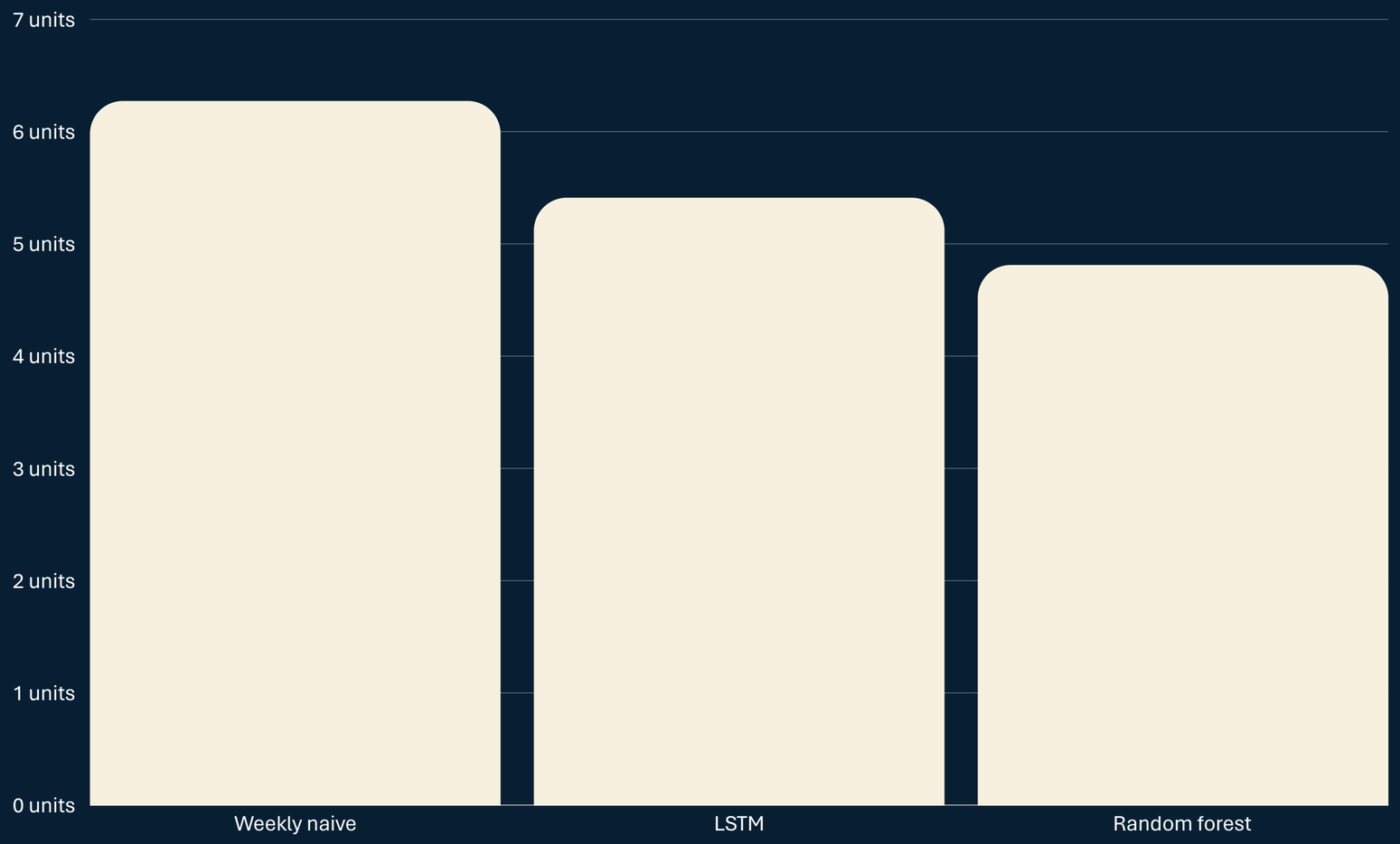
16 LSTM units

7-day output

Stop after 5 epochs without validation improvement · maximum 50

Better weekly totals — not the lowest daily MAE

The final test confirmed lower daily error



23.3%

lower daily MAE in 2017
6.273 → 4.812 units
Error reduction, not accuracy

LSTM still led on weekly-total MAE: 17.343

Worked example: an order of 38 units

$$77.52 + 10 - 50 = 37.52$$

5-day forecast buffer inventory position

ROUND UP

38 units

Available 50 · Outstanding 0 · Lead 2 days · Review 3 days · Buffer 10

How the text assistant understands a question



A confidence threshold rejects unrelated questions.

Live application demonstration



1 Select Product 1
2 Enter 50 · 0 · 2 · 3 · 10
3 Calculate → 38 units
4 Ask a forecast question

Historical demonstration estimates · January 2018



Forecast fairly. Order clearly.

Sunrise Mini Market is fictional. January 2018 outcomes are unknown. Real cost savings and stockout reductions have not been measured.

Questions?